

The Privacy Equilibrium Toolkit: Simulating Multi-User Negotiations of Augmented Reality Sensing Policies

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Abstract

This encore submission presents the Privacy Equilibrium toolkit [9], a research platform for simulating how multi-user negotiations of augmented reality (AR) sensing policies could be conducted in the future. Our overarching goal is to balance the user experience and privacy needs of multiple users and bystanders in shared spaces, an open challenge that is otherwise difficult to evaluate without widespread real-world usage of everyday AR. We discuss insights from evaluations with security & privacy experts, and reflect on opportunities for LLM-driven simulation to enable richer modeling of privacy risks.

CCS Concepts

- **Human-centered computing** → **Mixed / augmented reality**;
- **Security and privacy** → *Usability in security and privacy*.

Keywords

simulation, access control

1 Introduction

Wearable intelligent systems that enhance users’ everyday workflows, such as AI enabled smartglasses and augmented reality (AR) displays, raise novel privacy and safety concerns for both users and bystanders [2, 11]. Due to their reliance on continuous environmental and biometric input data, these systems can reveal people’s identities [5], behavioral patterns and health conditions [7], and sensitive details about their surroundings [13].

The security & privacy community has proposed approaches to regulate the use of continuous sensing at the interpersonal level (e.g., access control frameworks that restrict data capture for individuals users [4] or all those in a space [12]), as well as legal policies that go beyond protections for camera-based recording devices [1]. However, bridging the gap to real-world implementation is challenging: the everyday use of wearable AI systems is not yet widespread, leaving us without a comprehensive understanding of how such proposals would affect diverse users across contexts. For example, a world-driven approach to access control [12] (e.g., universally restricting camera use near sensitive locations) might exclude users who have a critical sensing need, such as those who use visual assistive technologies. Conversely, user-driven approaches [4] that let individuals define when and where their devices may sense can protect wearer privacy, but offer little recourse to bystanders.

Aligned with the PoliSim @ CHI 2026 workshop, we view simulation as a promising means of designing and testing future policy

for wearable AI systems, from conceptual formulation to generating requirements for implementation. As one example, **this short report presents our prior work on the Privacy Equilibrium toolkit [9], which enables exploring how to negotiate AR sensing policies** to balance user experience and privacy among multiple AR users and bystanders. We discuss the design and implementation of the toolkit, along with insights from evaluations with security & privacy experts.

2 The Privacy Equilibrium Toolkit

In our UIST 2025 paper [9], we developed a simulation and analysis toolkit¹ for HCI and security & privacy researchers to experiment with novel access control techniques that, beyond a single AR user, aim to meet the privacy needs of everyone co-located in a space. At the core is an optimization framework that adjusts access to AR sensing capabilities (e.g., spatial mapping, gesture and voice recognition) in a fine-grained manner to maximize privacy gain while minimizing UX loss, aiming to achieve an “equilibrium.” This section walks through three key processes embedded in our toolkit that support “triangulating” on possible negotiated policies [14].

- (1) **Specifying the Context of Use:** Sensing policies can have various implications depending on the physical and social context in which AR is used. To support evaluation across diverse settings, the toolkit supports defining usage scenarios in terms of individuals’ AR sensing requirements, privacy preferences, and movements within a physical space. We represent AR users and non-users as avatars, each assigned an *AR Sensing Profile* (i.e., which sensing capabilities their application requires to function at a high quality) and an *Interpersonal Privacy Profile* (i.e., sensing capabilities they are comfortable with others around them utilizing, based on privacy personas [3]). To provoke consideration of the social and privacy norms of different physical settings, the simulation engine renders these avatars within 3D floorplans (e.g., a hospital or office, as shown in Fig. 1A). Animated motion sequences simulate how people move through the space.
- (2) **Configuring AR Sensing Negotiations:** Next, researchers can experiment with different ways to negotiate sensing policies, which we model as a process of adjusting each user’s *AR Sensing Profile* to align with the *Interpersonal Privacy Profile* of co-located individuals. We consider two dimensions of negotiations that result in different equilibrium points between UX and privacy. First, the *Equilibrium Scope* defines the users involved:

This paper was prepared for PoliSim@CHI 2026: LLM Agent Simulation for Policy, a workshop at CHI 2026 (CHI Conference on Human Factors in Computing Systems), April 16, 2026, Barcelona, Spain.

¹**Video Demonstration of Privacy Equilibrium Toolkit:** <https://youtu.be/B631d4jS58c>

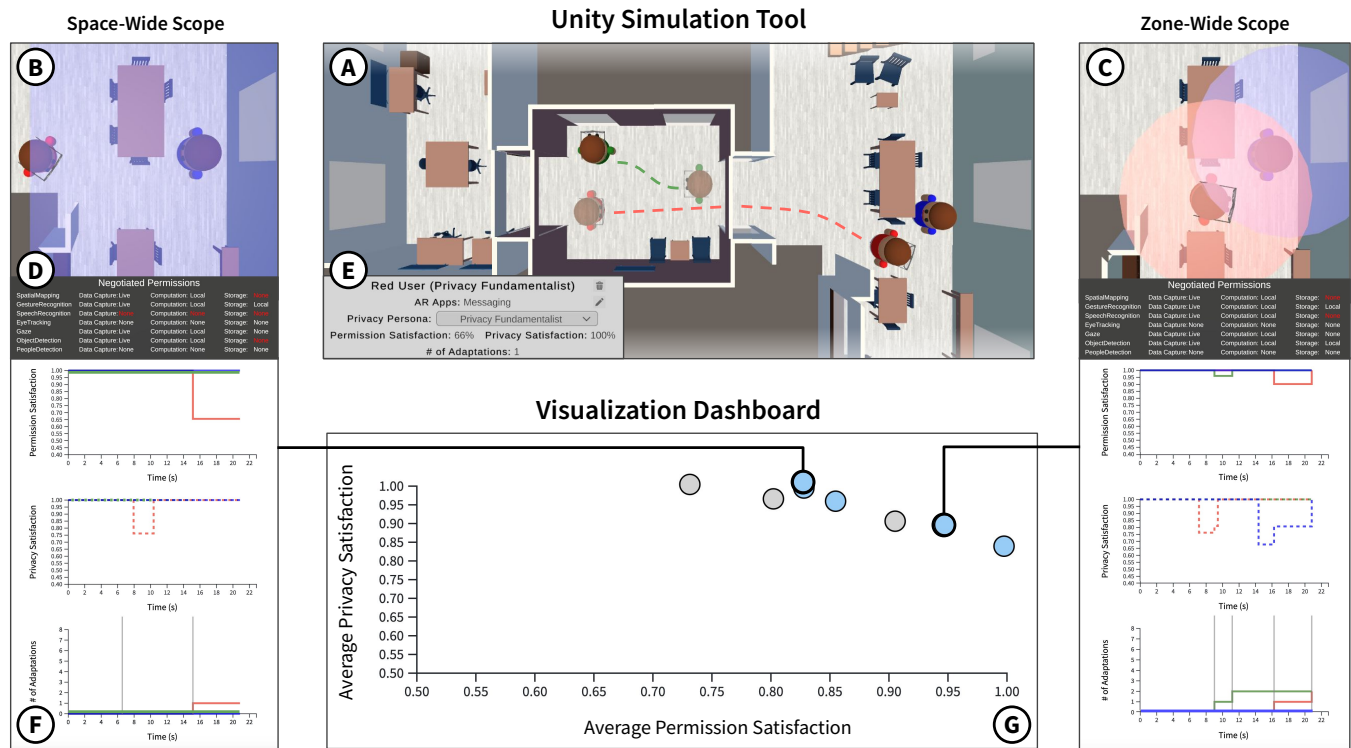


Figure 1: Privacy Equilibrium Toolkit Overview. Our toolkit includes a simulation engine implemented in Unity (A), which supports exploring AR usage behaviors and privacy risks across public environments, such as the office space shown here. Avatars representing users who leverage different AR applications and have varying privacy needs can be configured and animated to move around the space. Depending on the specified scope, sensing negotiations are triggered when users enter designated regions (Space-Wide; B) or come into proximity with other people (Zone-Wide, Peer-to-Peer; C). The simulation engine supports real-time analysis of negotiated sensing capabilities (D) and Equilibrium Scores (E). A web-based dashboard enables post-hoc analysis of score trends over time (F) and comparison of UX-privacy tradeoffs incurred by different Optimization Strategies, visualized along a Pareto curve (G).

all users in the physical setting (Space-Wide), in close proximity of each other (Zone-Wide), or pairs of users (Peer-to-Peer). Second, the *Optimization Strategy* defines whether more weight is given to satisfying the UX needs of all (Interaction-First), privacy needs of all (Privacy-First), or the sensing policy of a designated space owner (Ownership-Based). Based on users' movements and interactions with each other, sensing negotiations are dynamically triggered (Fig. 1B,C).

- (3) **Analyzing UX and Privacy Tradeoffs:** To help researchers understand to what extent different negotiation strategies bridge competing user goals, the toolkit offers two complementary analysis modes. In the simulation engine, real-time score displays approximate each user's levels of UX and privacy satisfaction (based on the proportion of permissions granted from their profiles). A web-based visualization dashboard supports post-hoc analysis through timeline plots, showing score trends across a full interaction session (Fig. 1F). The dashboard also plots Pareto curves that map different negotiation strategies

by their average UX and privacy satisfaction scores, highlighting the optimal (non-dominated) points that effectively balance tradeoffs (Fig. 1G).

3 Insights from Simulation Evaluations and Toolkit Walkthroughs with Security & Privacy Experts

To evaluate our framework’s expressiveness in balancing user needs, we conducted two studies. First, we ran application-driven simulations using our toolkit across two scenarios involving multi-user interactions in a hospital and office. We systematically varied the *Equilibrium Scope* and *Optimization Strategy* to assess how well different negotiation approaches resolved conflicting AR usage and bystander privacy requirements. We also conducted walkthroughs evaluations with security & privacy researchers, who used the toolkit to reason about the best negotiation strategy for both scenarios and suggest improvements to our optimization framework. Together, these studies surfaced three key themes.

- (1) **Benefits of flexible, fine-grained negotiation:** Experts emphasized the need for granular control over how and to whom

sensing restrictions are applied, and identified several framework components that support this. The Zone-Wide scope was perceived as offering the most flexibility, avoiding blanket restrictions on users who are not in proximity. Experts also valued the user control afforded by our optimization approach, noting that all users have some say in the negotiation. Our simulations similarly demonstrated that the Space-Wide, Interaction-First strategy consistently produced non-dominated points on the Pareto frontier, better balancing competing preferences than single user-driven access control approaches from prior work.

- (2) **Limitations in modeling the impact of sensing restrictions and severity of privacy exposure:** While experts could use the UX and privacy satisfaction scores to gauge strategy tradeoffs, they noted these metrics offer only a coarse-grained approximation. On the UX side, they wanted to understand how AR applications would adapt when sensing is restricted [10] to minimize adverse impacts for users with critical sensing needs (e.g., for accessibility). On the privacy side, experts stressed that a single score cannot capture the nuance of how users perceive privacy violations, or account for factors such as data retention policies and the likelihood of sensitive data exposure. As one expert noted, privacy loss is asymmetric: unlike UX, it cannot simply be recovered once incurred.
- (3) **Toolkit visualizations supported multi-level analysis of negotiation strategies:** The Unity simulation engine's motion paths, zone intersections, and in-situ score displays helped experts reason about both individual negotiations and broader patterns across a session. Visualizing when and where users entered each other's sensing range allowed experts to assess privacy risk based on proximity and device orientation, not just score values. The dashboard's timeline plots further enabled what one expert described as "local vs. global planning:" examining both the depth and duration of score dips within a negotiation, which can be indicative of fairness across the full interaction session. Experts suggested future improvements including first-person perspective views of AR users' sensing fields and embedded simulation video clips within the dashboard to support simultaneous local and global analysis.

4 Contributions to the Workshop

Through participating in the PoliSim Workshop at CHI 2026, we hope to share first-hand insights from designing, developing, and evaluating a simulation tool aimed at capturing holistic user goals. We also hope to contribute to broader discussions around the benefits and limitations of coarse versus fine-grained modeling approaches when it comes to informing future policy.

In turn, we hope to learn from other participants about LLM-driven techniques for simulating users with greater nuance and generalizability [8] (e.g., modeling dynamic privacy preferences across contexts [6], generating low and high-risk scenarios aligned with those preferences). We are also eager to discuss how to meaningfully involve key policymaking stakeholders, including strategies for sustaining partnerships and productive dialogues over time.

5 Authors' Backgrounds

Shwetha Rajaram is a PhD candidate at the University of Michigan, in the School of Information. Her research develops meaningful interactions with wearable intelligent systems (e.g., augmented reality) while safeguarding privacy.

Jiasi Chen is an Associate Professor at the University of Michigan, in the department of Electrical Engineering and Computer Science. Her research focus is on mobile computing and multimedia systems, including XR security and privacy.

Michael Nebeling is an Associate Professor at the University of Michigan, in the School of Information. He studies the design of usable and safe everyday, always-on AR experiences, with a particular interest in novel interaction techniques and tools to empower end-users.

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